

Evidence-Calibrated Natural-Language Explanations for Autonomous Navigation

Anonymous Submission

TNI-AIoT 2026 — Early-Career / Demonstration Abstract (non-archival)

Abstract—An autonomous robot can offer a physically correct explanation that its available evidence does not justify. We study *evidence-calibrated specificity*: robot language should become more informative as diagnostic evidence becomes available, while withholding unsupported causal refinements. Using the CRANE simulation platform, we formalize claim-specific evidence contracts, construct nested evidence interventions on fixed navigation episodes, and compare a tool-enabled language-model agent with a diagnostic-contract method combining maximal supported diagnosis, constrained realization, and verification. In exploratory automated rubric scoring of 511 retained paired land/Nav2 episodes, the agent had an adverse claim in 338 (66.1%; Wilson 95% CI 61.9–70.1%) and the contract method in none (0–0.75%). At the richest evidence level, descriptive essential-unit coverage was 0.897 versus 0.952. We evaluate evidential reliability alongside useful diagnostic information.

Index Terms—autonomous navigation, robot explanations, natural language, evidence grounding, diagnostic verification

I. WHY EVIDENCE-CALIBRATED ROBOT LANGUAGE?

Navigation explanations connect a robot’s execution record to an operator’s understanding of its decisions and failures. In AIoT deployments, this report links sensing and action to edge, cloud, and human consumers; its evidential limits must survive that communication. A command–response discrepancy can aid diagnosis; attributing it to motor failure requires evidence that distinguishes alternative causes. Even a correct hidden-cause attribution can exceed what the robot could justify.

Robot explanation systems already organize execution evidence: REFLECT summarizes experiences for failure explanation and correction [1], and HEXAR coordinates specialized component explainers [2]. Source attribution and atomic factuality also have established precedents [3], [4]. Our contribution is a scoped combination for navigation: testing whether explanatory specificity tracks controlled evidence availability while preserving useful partial diagnosis.

II. THREE CONTRIBUTIONS

C1: Evidence-calibrated explanation formalism. Atomic claims have evidence requirements and diagnostic abstraction levels. Contracts preserve supported partial diagnosis, ambiguity, limitations, and rejection of false failure premises. Evaluator-only physical truth is separate from robot-visible support: *physically true but unsupported* is an adverse category.

C2: Controlled navigation benchmark. Each physical episode is held fixed across removal-only masks: E0 provides outcome; E1 adds a source-qualified action/recovery trace; E2 adds delivered command; E3 adds synchronized odometry

TABLE I
EXPLORATORY RESULTS; DESCRIPTIVE ESSENTIAL-UNIT COVERAGE.

Measure	B2 agent	B4 contract
Any adverse claim	338/511	0/511
Wilson 95% interval	61.9–70.1%	0–0.75%
Accounting-complete sensitivity	325/494	0/494
Unsupported causal-claim flag	147/511	0/511
Contradicted-claim flag	43/511	0/511
Essential-unit coverage at E3	0.897	0.952

and governed diagnostic computation. We analyze persistent command–motion discrepancy and measured response recovery. Fixed episodes isolate evidence availability; masks and claims remain clustered within episodes.

C3: Diagnostic-contract method and comparison. B2 is a repository-aware language-model agent that uses tools to inspect evidence and source/configuration files, perform permitted diagnostic computations, and generate its answer. B4 computes the maximal supported diagnosis, selects permitted clauses, fills numeric slots through templates, and verifies the realized claims. Both receive the same relevant robot-visible evidence and access to the source/configuration and diagnostic primitives needed for information parity. B4 constrains the diagnosis-to-language boundary rather than granting access to hidden physical truth.

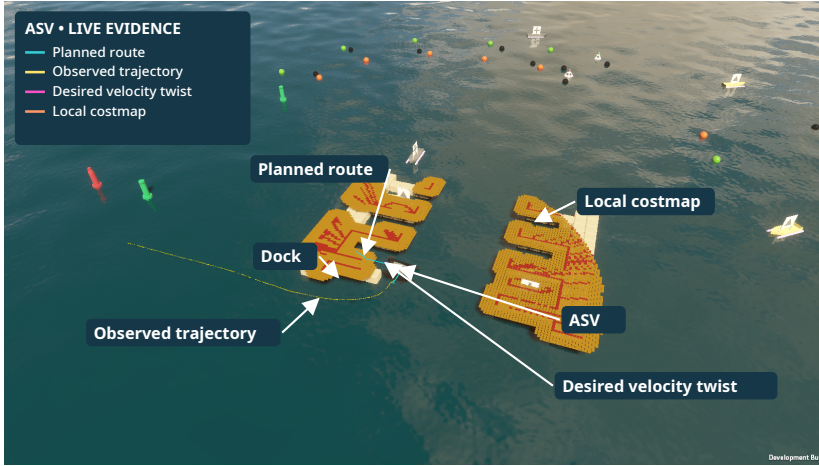
III. EVALUATION AND RETAINED RESULTS

A deterministic rubric extracts atomic claims from answer text, checks their evidence requirements against the visible level, and labels support, contradiction, insufficient evidence, physical truth without support, or uninterpretable content. It uses the same rules for both methods and does not read method identity or B4’s internal clauses. An episode is flagged if any substantive claim on its available ladder is adverse. Parity checks, retained technical failures, removal-only masks, and deterministic reruns support traceability. The independent analysis unit is the episode, never the mask or claim.

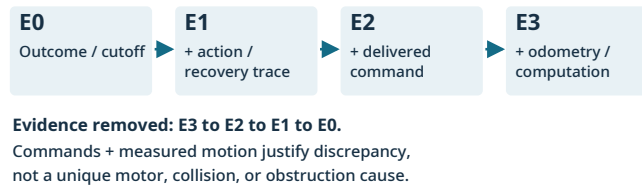
Collection stopped before its first confirmatory look and blinded annotation. This post-stop analysis is exploratory automated scoring. We report the 511 episodes with at least one paired level and the 494 accounting-complete sensitivity; denominators differ by level (Fig. 1).

The observed adverse-rate difference is –66.1 percent-age points (B4 minus B2). B2’s adverse labels include 420 insufficient-evidence claims, 45 contradiction flags, and four

(a) CRANE platform: ASV navigation evidence



(b) Same episode, nested evidence



One evidence boundary; two explainers

B2 • tool-enabled LLM agent

Visible evidence + source/configuration
Permitted diagnostic computations
Natural-language answer returned verbatim

B4 • diagnostic-contract method

Same visible evidence and source access
Maximal supported diagnosis
Constrained clauses + exact numeric slots
Final claim verification

ASV illustration; results below: land/Nav2.
Physical truth is evaluator-only.

(c) Episodes flagged by evidence level

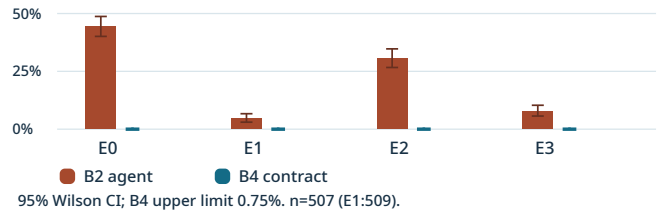


Fig. 1. CRANE evidence interface and evaluation. (a) ASV demonstration with recorded route, trajectory, desired velocity twist, and costmap; white arrows identify scene features. (b) Fixed land episode under removal-only evidence conditions. (c) B2/B4 adverse-claim flags with marginal Wilson 95% intervals; levels repeat within episodes. The ASV illustration is outside the quantitative land cohort.

physical-truth-without-support labels. B4 coverage is lower at E0/E1 and higher at E2/E3: the result is a specificity–coverage tradeoff, with useful detail retained at the richest level, rather than uniform caution.

IV. IMPLICATIONS AND BOUNDARIES

A platform for realistic navigation evidence. CRANE (Cross-domain Autonomous Robotics Navigation Engine) is the study’s simulation platform. Unity/PhysX dynamics and sensor interfaces connect to ROS navigation execution in land and ASV environments. Shared interfaces support cross-domain investigation in realistic scenarios; quantitative evidence here remains land/Nav2 simulation.

Useful diagnosis with explicit causal boundaries. At E2, a delivered velocity command does not establish how the robot moved. At E3, the synchronized comparison can justify “commanded motion was not reflected in measured motion,” while leaving motor failure, slip, and obstruction unresolved. Similarly, a measured response recovery need not establish task success. Partial diagnosis preserves useful observations. The ASV demonstration likewise separates software success from sustained physical docking.

Claim-level checks provide a bounded reflection step: before reporting an explanation, the system asks what each claim requires and whether the execution record supplies it. This directly addresses TNI-AIoT themes of reliable autonomous agents, explainability and monitoring, dependable sensing-to-

action processes, and reproducible benchmarks. The evidence contract offers robotics researchers an auditable interface between navigation diagnostics and natural-language expression.

The findings are simulation-only and rubric-relative. Template realization and scoring share an ontology, so B4’s zero flagged rate is not proof of zero semantic error. Automated scoring is not human validation; coverage measures communicated contract units, not operator usefulness. One agent configuration cannot establish superiority over all agents. Independent semantic assessment, prospective replication, and operator evaluation are the next tests of whether these evidence boundaries improve real-world navigation explanation.

REFERENCES

- [1] Z. Liu, A. Bahety, and S. Song, “REFLECT: Summarizing robot experiences for failure explanation and correction,” in *Proc. CoRL*, PMLR 229, pp. 3468–3484, 2023.
- [2] T. Love *et al.*, “HEXAR: A hierarchical explainability architecture for robots,” arXiv:2601.03070, 2026.
- [3] H. Rashkin *et al.*, “Measuring attribution in natural language generation models,” *Comput. Linguist.*, vol. 49, no. 4, pp. 777–840, 2023.
- [4] S. Min *et al.*, “FACTScore: Fine-grained atomic evaluation of factual precision in long form text generation,” in *Proc. EMNLP*, pp. 12076–12100, 2023.